

Hudson Dalby

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EDUCATION

Master of Science: Computer Engineering with Deep Learning Certificate
Bachelor of Science: Computer Engineering
University of Utah, John & Marcia Price College of Engineering

Salt Lake City, UT
Anticipated December 2027
Concurrent

Related Coursework

Embedded Systems, Computer Architecture, Digital & Computer Systems, Circuits & Systems, Deep Learning, Robotics

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Verilog

Embedded/Hardware: STM32, FPGA (Intel Quartus), PCB Design, Arduino, Raspberry Pi

Tools & ML: Linux, GDB, MATLAB, PyTorch, NumPy

TECHNICAL PROJECTS

Motor Driver PCB – KiCad, C

January 2026 – March 2026

- Designed and implemented a 6V DC motor driver circuit using an H-bridge topology for bidirectional control.
- Developed a custom PCB layout (schematic + routing) in CAD, optimized for signal integrity.
- Implemented closed-loop PID motor control in C on an STM32 microcontroller using encoder feedback.
- Validated circuit performance using oscilloscope measurements of switching signals and motor response.

xv6 OS Extensions – C, x86-64, Linux

February 2026 – Present

- Extended the xv6 kernel across boot, program loading, and user-space execution.
- Implemented a Unix-like shell with job control, including foreground/background execution and signal handling.
- Developed an ELF loader to map program segments into memory, handle relocations, and transfer control.
- Analyzed and modified the boot process and virtual memory setup, including early page table initialization and address space creating.

FPGA Computer – Verilog, Assembly, Intel Quartus

September 2025 – December 2025

- Co-designed a custom RISC-inspired CPU on an Intel FPGA using Verilog.
- Implemented core CPU components including datapath, control logic, ALU, register file, and memory interfaces.
- Developed a custom ISA and wrote assembly-level software, including a real-time game application.
- Integrated VGA graphics output and external game controller input via memory mapped I/O.

EXPERIENCE

Domino's Pizza

Riverton, UT

Delivery Driver

June 2023 – August 2024

- Ensured a positive customer experience for customers and made deliveries in a timely manner.
- Responsible for handling cash and credit transactions.

Walmart Neighborhood Market

Riverton, UT

Online Order Fulfillment

January 2022 – May 2023

- Ensured accurate fulfillment and quick dispensing of online orders.
- Responsible for organization of pending orders, merchandise stocking and inventory.